

AP[®] Calculus AB Practice Exam

From the 2014 Administration

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A A

SECTION I, Part A

Time—55 minutes

Number of questions—28

A CALCULATOR MAY NOT BE USED ON THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in the exam book. Do not spend too much time on any one problem.

In this exam:

- (1) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.
- (2) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix “arc” (e.g., $\sin^{-1} x = \arcsin x$).

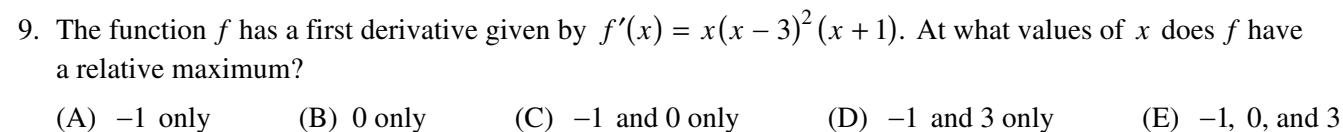
1. $\int_2^x (3t^2 - 1) dt =$

- 4-

7. $\lim_{x \rightarrow \infty} \frac{x^3}{e^{3x}}$ is

8. Using the substitution $u = \sin(2x)$, $\int_{\pi/6}^{\pi/2} \sin^5(2x)\cos(2x) dx$ is equivalent to

- 8-



10. Let f be the function defined above. For what value of b is f continuous at $x = 2$?

(A) -3 (B) $\sqrt{2}$ (C) 3 (D) 5 (E) There is no such value of b .

13. $\frac{d}{dx}\left(\frac{x+1}{x^2+1}\right) =$

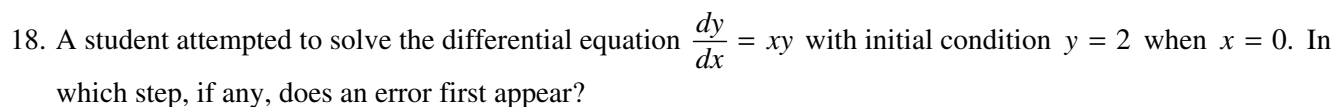
(A) $\frac{x^2 + 2x - 1}{(x^2 + 1)^2}$

(B) $\frac{-x^2 - 2x + 1}{x^2 + 1}$

(C) $\frac{-x^2 - 2x + 1}{(x^2 + 1)^2}$

(D) $\frac{3x^2 + 2x + 1}{(x^2 + 1)^2}$

(E) $\frac{1}{2x}$



Step 2: $\ln |y| = \frac{x^2}{2} + C$

Step 4: Since $y = 2$ when $x = 0$, $2 = e^0 + C$.

Step 5: $y = e^{x^2/2} + 1$

- (A) Step 2
(B) Step 3
(C) Step 4
(D) Step 5
(E) There is no error in the solution.

19. For what values of x does the graph of $y = 3x^5 + 10x^4$ have a point of inflection?

- (A) $x = -\frac{8}{3}$ only
- (B) $x = -2$ only
- (C) $x = 0$ only
- (D) $x = 0$ and $x = -\frac{8}{3}$
- (E) $x = 0$ and $x = -2$

20. $\lim_{x \rightarrow 2} \frac{\ln(x+3) - \ln(5)}{x-2}$ is

- (A) 0 (B) $\frac{1}{5}$ (C) $\frac{1}{2}$ (D) 1 (E) nonexistent



$$f(x) = \begin{cases} 3x + 5 & \text{when } x < -1 \\ -x^2 + 3 & \text{when } x \geq -1 \end{cases}$$

23. If f is the function defined above, then $f'(-1)$ is

- (A) -3 (B) -2 (C) 2 (D) 3 (E) nonexistent

24. Let f be the function defined by $f(x) = \frac{(3x+8)(5-4x)}{(2x+1)^2}$. Which of the following is a horizontal asymptote to the graph of f ?

- (A) $y = -6$
(B) $y = -3$
(C) $y = -\frac{1}{2}$
(D) $y = 0$
(E) $y = \frac{3}{2}$

$$(A) \quad \frac{dy}{dx} = xy - x$$

(B) $\frac{dy}{dx} = xy + x$

(C) $\frac{dy}{dx} = y - x^2$

(D) $\frac{dy}{dx} = (y - 1)x^2$

(E) $\frac{dy}{dx} = (y - 1)^3$

DO NOT GO ON TO PART B UNTIL YOU ARE TOLD TO DO SO.

B**B****B****B****B****B****B****B****B****CALCULUS AB****SECTION I, Part B****Time—50 minutes****Number of questions—17**

A GRAPHING CALCULATOR IS REQUIRED FOR SOME QUESTIONS ON
THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in the exam book. Do not spend too much time on any one problem.

BE SURE YOU ARE USING PAGE 3 OF THE ANSWER SHEET TO RECORD YOUR ANSWERS TO QUESTIONS NUMBERED 76–92.

YOU MAY NOT RETURN TO PAGE 2 OF THE ANSWER SHEET.

In this exam:

- (1) The exact numerical value of the correct answer does not always appear among the choices given. When this happens, select from among the choices the number that best approximates the exact numerical value.
- (2) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.
- (3) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix “arc” (e.g., $\sin^{-1}x = \arcsin x$).

B**B****B****B****B****B****B****B****B**

76. A particle moves along a straight line so that at time $t > 0$ the position of the particle is given by $s(t)$, the velocity is given by $v(t)$, and the acceleration is given by $a(t)$. Which of the following expressions gives the average velocity of the particle on the interval $[2, 8]$?

(A) $\frac{1}{6} \int_2^8 a(t) dt$

(B) $\frac{1}{6} \int_2^8 s(t) dt$

(C) $\frac{s(8) - s(2)}{6}$

(D) $\frac{v(8) - v(2)}{6}$

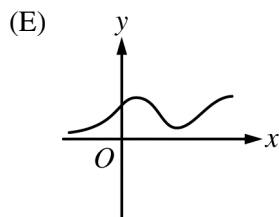
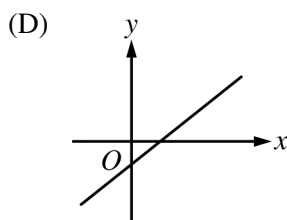
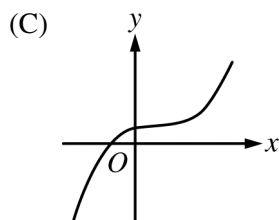
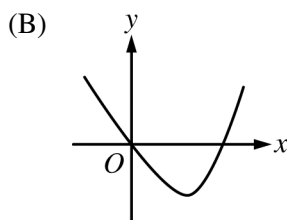
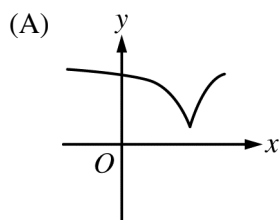
(E) $v(8) - v(2)$

-
77. If $\sin\left(\frac{1}{x^2 + 1}\right)$ is an antiderivative for $f(x)$, then $\int_1^2 f(x) dx =$

(A) -0.281 (B) -0.102 (C) 0.102 (D) 0.260 (E) 0.282

B**B****B****B****B****B****B****B****B**

78. The function f is differentiable and increasing for all real numbers x , and the graph of f has exactly one point of inflection. Of the following, which could be the graph of f' , the derivative of f ?



B**B****B****B****B****B****B****B****B**

79. A vase has the shape obtained by revolving the curve $y = 2 + \sin x$ from $x = 0$ to $x = 5$ about the x -axis, where x and y are measured in inches. What is the volume, in cubic inches, of the vase?

(A) 10.716 (B) 25.501 (C) 33.666 (D) 71.113 (E) 80.115

x	$f(x)$
1	2.4
3	3.6
5	5.4

80. The table above gives selected values of a function f . The function is twice differentiable with $f''(x) > 0$. Which of the following could be the value of $f'(3)$?

(A) 0.6 (B) 0.7 (C) 0.9 (D) 1.2 (E) 1.5

B**B****B****B****B****B****B****B****B**

81. At time $t = 0$ years, a forest preserve has a population of 1500 deer. If the rate of growth of the population is modeled by $R(t) = 2000e^{0.23t}$ deer per year, what is the population at time $t = 3$?

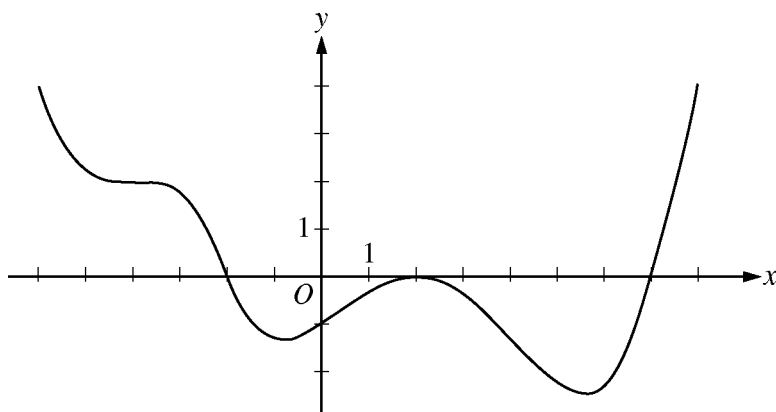
(A) 3987

(B) 5487

(C) 8641

(D) 10,141

(E) 12,628

B**B****B****B****B****B****B****B****B**

Graph of f'

82. The figure above shows the graph of f' , the derivative of function f , for $-6 < x < 8$. Of the following, which best describes the graph of f on the same interval?
- (A) 1 relative minimum, 1 relative maximum, and 3 points of inflection
 - (B) 1 relative minimum, 1 relative maximum, and 4 points of inflection
 - (C) 2 relative minima, 1 relative maximum, and 2 points of inflection
 - (D) 2 relative minima, 1 relative maximum, and 4 points of inflection
 - (E) 2 relative minima, 2 relative maxima, and 3 points of inflection

B**B****B****B****B****B****B****B****B**

83. Let f and g be continuous functions such that $\int_0^6 f(x) \, dx = 9$, $\int_3^6 f(x) \, dx = 5$, and $\int_3^0 g(x) \, dx = -7$. What is

the value of $\int_0^3 \left(\frac{1}{2} f(x) - 3g(x) \right) dx$?

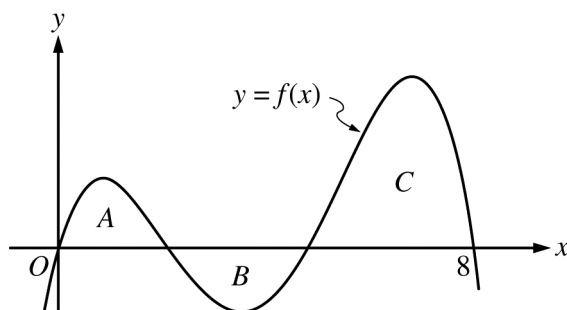
(A) -23

(B) -19

(C) $-\frac{17}{2}$

(D) 19

(E) 23

B**B****B****B****B****B****B****B****B**

84. The regions A , B , and C in the figure above are bounded by the graph of the function f and the x -axis. The area of region A is 14, the area of region B is 16, and the area of region C is 50. What is the average value of f on the interval $[0, 8]$?

(A) 6 (B) 10 (C) $\frac{40}{3}$ (D) $\frac{80}{3}$ (E) 48

-
85. A particle moves along the x -axis so that its velocity at time $t \geq 0$ is given by $v(t) = \frac{t^2 - 1}{t^2 + 1}$. What is the total distance traveled by the particle from $t = 0$ to $t = 2$?

(A) 0.214 (B) 0.320 (C) 0.600 (D) 0.927 (E) 1.600

B**B****B****B****B****B****B****B****B**

86. Line ℓ is tangent to the graph of $y = e^x$ at the point (k, e^k) . What is the positive value of k for which the y -intercept of ℓ is $\frac{1}{2}$?

(A) 0.405

(B) 0.768

(C) 1.500

(D) 1.560

(E) There is no such value of k .

87. A differentiable function f has the property that $f'(x) \leq 3$ for $1 \leq x \leq 8$ and $f(5) = 6$. Which of the following could be true?

I. $f(2) = 0$

II. $f(6) = -2$

III. $f(7) = 13$

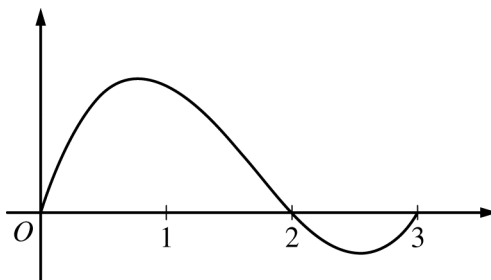
(A) I only

(B) II only

(C) I and II only

(D) I and III only

(E) II and III only

B**B****B****B****B****B****B****B****B**

Graph of f

88. The graph of the differentiable function f is shown in the figure above. Let h be the function defined by $h(x) = \int_0^x f(t) dt$. Which of the following correctly orders $h(2)$, $h'(2)$, and $h''(2)$?

- (A) $h(2) < h'(2) < h''(2)$
- (B) $h'(2) < h(2) < h''(2)$
- (C) $h'(2) < h''(2) < h(2)$
- (D) $h''(2) < h(2) < h'(2)$
- (E) $h''(2) < h'(2) < h(2)$

B**B****B****B****B****B****B****B****B**

89. What is the area of the region enclosed by the graphs of $y = e^x - 2$, $y = \sin x$, and $x = 0$?

- (A) 0.239 (B) 0.506 (C) 0.745 (D) 2.340 (E) 3.472

90. A particle moves along a line so that its velocity is given by $v(t) = -t^3 + 2t^2 + 2^{-t}$ for $t \geq 0$. For what values of t is the speed of the particle increasing?

- (A) $(0, 0.177)$ and $(1.256, \infty)$
(B) $(0, 1.256)$ only
(C) $(0, 2.057)$ only
(D) $(0.177, 1.256)$ only
(E) $(0.177, 1.256)$ and $(2.057, \infty)$

B**B****B****B****B****B****B****B****B**

91. Let F be a function defined for all real numbers x such that $F'(x) > 0$ and $F''(x) > 0$. Which of the following could be a table of values for F ?

(A)

x	$F(x)$
1	-3
2	-4
3	-6
4	-9

(B)

x	$F(x)$
1	-3
2	-1
3	3
4	19

(C)

x	$F(x)$
1	-3
2	0
3	3
4	6

(D)

x	$F(x)$
1	-3
2	5
3	11
4	13

(E)

x	$F(x)$
1	-3
2	-4
3	-3
4	0

B**B****B****B****B****B****B****B****B**

x	$f(x)$	$g(x)$	$f'(x)$
-4	0	-9	5
-2	4	-7	4
0	6	-4	2
2	7	-3	1
4	10	-2	3

92. The table above gives values of the differentiable functions f and g , and f' , the derivative of f , at selected values of x . If $g(x) = f^{-1}(x)$, what is the value of $g'(4)$?

(A) $-\frac{1}{3}$ (B) $-\frac{1}{4}$ (C) $-\frac{3}{100}$ (D) $\frac{1}{4}$ (E) $\frac{1}{3}$

CALCULUS AB
SECTION II, Part A
Time—30 minutes
Number of problems—2

A graphing calculator is required for these problems.

GO ON TO THE NEXT PAGE.

t (hours)	0	0.4	0.8	1.2	1.6	2.0	2.4
$v(t)$ (miles per hour)	0	11.8	9.5	17.2	16.3	16.8	20.1

1. Ruth rode her bicycle on a straight trail. She recorded her velocity $v(t)$, in miles per hour, for selected values of t over the interval $0 \leq t \leq 2.4$ hours, as shown in the table above. For $0 < t \leq 2.4$, $v(t) > 0$.
- (a) Use the data in the table to approximate Ruth's acceleration at time $t = 1.4$ hours. Show the computations that lead to your answer. Indicate units of measure.

-
- (b) Using correct units, interpret the meaning of $\int_0^{2.4} v(t) dt$ in the context of the problem. Approximate $\int_0^{2.4} v(t) dt$ using a midpoint Riemann sum with three subintervals of equal length and values from the table.

1**1****1****1****1****1****1****1****1****1**

- (c) For $0 \leq t \leq 2.4$ hours, Ruth's velocity can be modeled by the function g given by $g(t) = \frac{24t + 5\sin(6t)}{t + 0.7}$. According to the model, what was Ruth's average velocity during the time interval $0 \leq t \leq 2.4$?

-
- (d) According to the model given in part (c), is Ruth's speed increasing or decreasing at time $t = 1.3$? Give a reason for your answer.

2. A store is having a 12-hour sale. The total number of shoppers who have entered the store t hours after the sale begins is modeled by the function S defined by $S(t) = 0.5t^4 - 16t^3 + 144t^2$ for $0 \leq t \leq 12$. At time $t = 0$, when the sale begins, there are no shoppers in the store.

(a) At what rate are shoppers entering the store 3 hours after the start of the sale?

-
- (b) Find the value of $\frac{1}{3} \int_6^9 S'(t) dt$. Using correct units, explain the meaning of $\frac{1}{3} \int_6^9 S'(t) dt$ in the context of this problem.

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- (c) The rate at which shoppers leave the store, measured in shoppers per hour, is modeled by the function L defined by $L(t) = -80 + \frac{4400}{t^2 - 14t + 55}$ for $0 \leq t \leq 12$. According to the model, how many shoppers are in the store at the end of the sale (time $t = 12$)? Give your answer to the nearest whole number.

-
- (d) Using the given models, find the time t , $0 \leq t \leq 12$, at which the number of shoppers in the store is the greatest. Justify your answer.

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CALCULUS AB
SECTION II, Part B
Time—60 minutes
Number of problems—4

No calculator is allowed for these problems.

DO NOT BREAK THE SEALS UNTIL YOU ARE TOLD TO DO SO.

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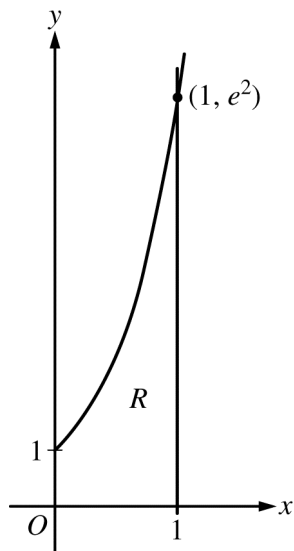
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NO CALCULATOR ALLOWED



3. Let $f(x) = e^{2x}$. Let R be the region in the first quadrant bounded by the graph of $y = f(x)$ and the vertical line $x = 1$, as shown in the figure above.
- (a) Write an equation for the line tangent to the graph of f at $x = 1$.

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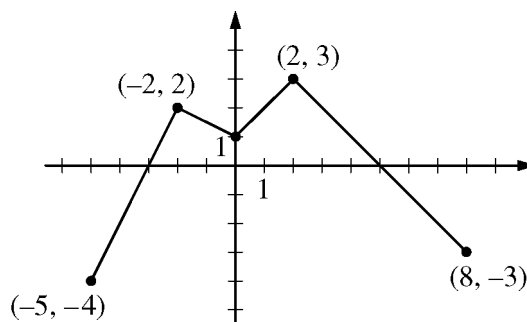
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NO CALCULATOR ALLOWED

(b) Find the area of R .

(c) Region R forms the base of a solid whose cross sections perpendicular to the y -axis are squares. Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid.

NO CALCULATOR ALLOWED

Graph of f

4. The continuous function f is defined on the interval $-5 \leq x \leq 8$. The graph of f , which consists of four line segments, is shown in the figure above. Let g be the function given by $g(x) = 2x + \int_{-2}^x f(t) dt$.

(a) Find $g(0)$ and $g(-5)$.

(b) Find $g'(x)$ in terms of $f(x)$. For each of $g''(4)$ and $g''(-2)$, find the value or state that it does not exist.

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NO CALCULATOR ALLOWED

(c) On what intervals, if any, is the graph of g concave down? Give a reason for your answer.

(d) The function h is given by $h(x) = g(x^3 + 1)$. Find $h'(1)$. Show the work that leads to your answer.

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5**5****5****5****5****5****5****5****5****5****NO CALCULATOR ALLOWED**

5. Particle X moves along the positive x -axis so that its position at time $t \geq 0$ is given by $x(t) = 5t^3 - 9t^2 + 7$.

(a) Is particle X moving toward the left or toward the right at time $t = 1$? Give a reason for your answer.

(b) At what time $t \geq 0$ is particle X farthest to the left? Justify your answer.

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NO CALCULATOR ALLOWED

- (c) A second particle, Y , moves along the positive y -axis so that its position at time t is given by $y(t) = 7t + 3$. At any time t , $t \geq 0$, the origin and the positions of the particles X and Y are the vertices of a triangle in the first quadrant. Find the rate of change of the area of the triangle at time $t = 1$. Show the work that leads to your answer.

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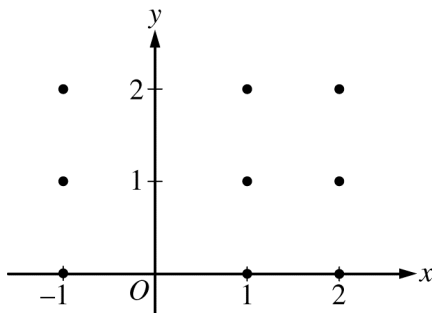
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NO CALCULATOR ALLOWED

6. Consider the differential equation $\frac{dy}{dx} = \left(1 - \frac{2}{x^2}\right)(y - 1)$, where $x \neq 0$. Let $y = f(x)$ be the particular solution to the differential equation with initial condition $f(1) = 2$.

(a) Find the slope of the line tangent to the graph of f at the point $(1, 2)$.

(b) On the axes provided, sketch a slope field for the given differential equation at the nine points indicated.



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6**6****6****6****6****6****6****6****6****6****NO CALCULATOR ALLOWED**

- (c) Find the particular solution $y = f(x)$ to the differential equation $\frac{dy}{dx} = \left(1 - \frac{2}{x^2}\right)(y - 1)$ with initial condition $f(1) = 2$.

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Answer Key for AP Calculus AB Practice Exam, Section I

Question 1: A	Question 24: B
Question 2: B	Question 25: D
Question 3: D	Question 26: A
Question 4: E	Question 27: D
Question 5: C	Question 28: A
Question 6: E	Question 76: C
Question 7: A	Question 77: A
Question 8: D	Question 78: A
Question 9: A	Question 79: E
Question 10: E	Question 80: B
Question 11: D	Question 81: D
Question 12: C	Question 82: A
Question 13: C	Question 83: B
Question 14: D	Question 84: A
Question 15: A	Question 85: D
Question 16: B	Question 86: B
Question 17: E	Question 87: C
Question 18: B	Question 88: E
Question 19: B	Question 89: C
Question 20: B	Question 90: E
Question 21: C	Question 91: B
Question 22: D	Question 92: D
Question 23: E	

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Question 1

t (hours)	0	0.4	0.8	1.2	1.6	2.0	2.4
$v(t)$ (miles per hour)	0	11.8	9.5	17.2	16.3	16.8	20.1

Ruth rode her bicycle on a straight trail. She recorded her velocity $v(t)$, in miles per hour, for selected values of t over the interval $0 \leq t \leq 2.4$ hours, as shown in the table above. For $0 < t \leq 2.4$, $v(t) > 0$.

- (a) Use the data in the table to approximate Ruth's acceleration at time $t = 1.4$ hours. Show the computations that lead to your answer. Indicate units of measure.
- (b) Using correct units, interpret the meaning of $\int_0^{2.4} v(t) dt$ in the context of the problem. Approximate $\int_0^{2.4} v(t) dt$ using a midpoint Riemann sum with three subintervals of equal length and values from the table.
- (c) For $0 \leq t \leq 2.4$ hours, Ruth's velocity can be modeled by the function g given by $g(t) = \frac{24t + 5\sin(6t)}{t + 0.7}$. According to the model, what was Ruth's average velocity during the time interval $0 \leq t \leq 2.4$?
- (d) According to the model given in part (c), is Ruth's speed increasing or decreasing at time $t = 1.3$? Give a reason for your answer.

(a) $a(1.4) \approx \frac{v(1.6) - v(1.2)}{1.6 - 1.2} = \frac{16.3 - 17.2}{1.6 - 1.2} = -2.25 \text{ miles/hr}^2$

2 : $\begin{cases} 1 : \text{approximation} \\ 1 : \text{units} \end{cases}$

- (b) $\int_0^{2.4} v(t) dt$ is the total distance Ruth traveled, in miles, from time $t = 0$ to time $t = 2.4$ hours.

$$\int_0^{2.4} v(t) dt \approx (0.8)(11.8) + (0.8)(17.2) + (0.8)(16.8) \\ = 36.64 \text{ miles}$$

3 : $\begin{cases} 1 : \text{interpretation} \\ 1 : \text{midpoint Riemann sum} \\ 1 : \text{approximation} \end{cases}$

(c) Average velocity $= \frac{1}{2.4} \int_0^{2.4} g(t) dt = 14.064 \text{ miles/hr}$

2 : $\begin{cases} 1 : \text{integral} \\ 1 : \text{answer} \end{cases}$

(d) Velocity $= g(1.3) = 18.096358 > 0$
Acceleration $= g'(1.3) = 3.761152 > 0$

2 : conclusion with reason

Ruth's speed is increasing at time $t = 1.3$ since velocity and acceleration have the same sign at this time.

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Question 2

A store is having a 12-hour sale. The total number of shoppers who have entered the store t hours after the sale begins is modeled by the function S defined by $S(t) = 0.5t^4 - 16t^3 + 144t^2$ for $0 \leq t \leq 12$. At time $t = 0$, when the sale begins, there are no shoppers in the store.

- (a) At what rate are shoppers entering the store 3 hours after the start of the sale?
- (b) Find the value of $\frac{1}{3} \int_6^9 S'(t) dt$. Using correct units, explain the meaning of $\frac{1}{3} \int_6^9 S'(t) dt$ in the context of this problem.
- (c) The rate at which shoppers leave the store, measured in shoppers per hour, is modeled by the function L defined by $L(t) = -80 + \frac{4400}{t^2 - 14t + 55}$ for $0 \leq t \leq 12$. According to the model, how many shoppers are in the store at the end of the sale (time $t = 12$)? Give your answer to the nearest whole number.
- (d) Using the given models, find the time t , $0 \leq t \leq 12$, at which the number of shoppers in the store is the greatest. Justify your answer.

(a) $S'(3) = 486$ shoppers/hour

(b)
$$\frac{1}{3} \int_6^9 S'(t) dt = \frac{1}{3}(S(9) - S(6))$$

$$= \frac{1}{3}(3280.5 - 2376) = 301.5$$

$$\frac{1}{3} \int_6^9 S'(t) dt \text{ is the average rate at which shoppers are entering}$$

$$\text{the store in shoppers/hr between times } t = 6 \text{ and } t = 9 \text{ hours.}$$

(c) $S(12) - \int_0^{12} L(t) dt = 195.701684$
 Therefore, there are approximately 196 shoppers in the store at the end of the sale.

(d) The number of shoppers in the store at time t is given by the function N defined by

$$N(t) = S(t) - \int_0^t L(x) dx$$

$$N'(t) = S'(t) - L(t) = 0 \text{ when } t = 5.545066$$

t	$N(t)$
0	0
5.545066	1361.832842
12	195.702

The number of shoppers in the store is greatest at time $t = 5.545$ hours.

1 : answer

2 :
$$\begin{cases} 1 : \text{value of } \frac{1}{3} \int_6^9 S'(t) dt \\ 1 : \text{meaning of } \frac{1}{3} \int_6^9 S'(t) dt \end{cases}$$

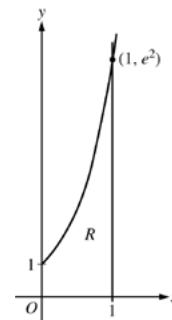
3 :
$$\begin{cases} 1 : \text{integral} \\ 1 : \text{uses } S(12) \\ 1 : \text{answer} \end{cases}$$

3 :
$$\begin{cases} 1 : \text{considers } S'(t) - L(t) = 0 \\ 1 : \text{answer} \\ 1 : \text{justification} \end{cases}$$

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Question 3

Let $f(x) = e^{2x}$. Let R be the region in the first quadrant bounded by the graph of $y = f(x)$ and the vertical line $x = 1$, as shown in the figure above.



- (a) Write an equation for the line tangent to the graph of f at $x = 1$.
- (b) Find the area of R .
- (c) Region R forms the base of a solid whose cross sections perpendicular to the y -axis are squares. Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid.

- (a) $f(1) = e^2$
 $f'(x) = 2e^{2x} \Rightarrow f'(1) = 2e^2$
 An equation for the tangent line is $y = e^2 + 2e^2(x - 1)$.

2 : $\begin{cases} 1 : f'(1) \\ 1 : \text{answer} \end{cases}$

- (b) $\text{Area} = \int_0^1 e^{2x} dx = \left[\frac{1}{2} e^{2x} \right]_{x=0}^{x=1} = \frac{1}{2}(e^2 - 1)$

3 : $\begin{cases} 1 : \text{integral} \\ 1 : \text{antiderivative} \\ 1 : \text{answer} \end{cases}$

- (c) $\text{Volume} = 1 + \int_1^{e^2} \left(1 - \frac{1}{2} \ln y \right)^2 dy$

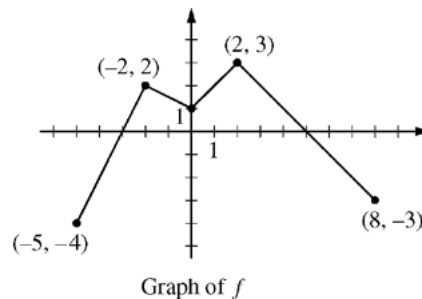
4 : $\begin{cases} 2 : \text{integrand} \\ 1 : \text{limits} \\ 1 : \text{answer} \end{cases}$

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Question 4

The continuous function f is defined on the interval $-5 \leq x \leq 8$. The graph of f , which consists of four line segments, is shown in the figure above.

Let g be the function given by $g(x) = 2x + \int_{-2}^x f(t) \, dt$.



- (a) Find $g(0)$ and $g(-5)$.
- (b) Find $g'(x)$ in terms of $f(x)$. For each of $g''(4)$ and $g''(-2)$, find the value or state that it does not exist.
- (c) On what intervals, if any, is the graph of g concave down? Give a reason for your answer.
- (d) The function h is given by $h(x) = g(x^3 + 1)$. Find $h'(1)$. Show the work that leads to your answer.

(a) $g(0) = 2 \cdot 0 + \int_{-2}^0 f(t) \, dt = 3$

$$g(-5) = 2 \cdot (-5) + \int_{-2}^{-5} f(t) \, dt = -10 + 3 = -7$$

(b) $g'(x) = 2 + f(x)$
 $g''(x) = f'(x)$

$$g''(4) = f'(4) = -1$$

$$g''(-2) = f'(-2) \text{ does not exist.}$$

- (c) The graph of g is concave down on the intervals $(-2, 0)$ and $(2, 8)$ since $g'(x) = 2 + f(x)$ decreases on those intervals.

(d) $h'(x) = g'(x^3 + 1) \cdot 3x^2$

$$\begin{aligned} h'(1) &= g'(2) \cdot 3 = (2 + f(2)) \cdot 3 \\ &= (2 + 3) \cdot 3 = 15 \end{aligned}$$

$$2 : \begin{cases} 1 : g(0) \\ 1 : g(-5) \end{cases}$$

$$3 : \begin{cases} 1 : g'(x) \\ 1 : g''(4) \\ 1 : g''(-2) \text{ does not exist} \end{cases}$$

1 : intervals and reason

$$3 : \begin{cases} 2 : \text{chain rule} \\ 1 : \text{answer} \end{cases}$$

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Question 5

Particle X moves along the positive x -axis so that its position at time $t \geq 0$ is given by $x(t) = 5t^3 - 9t^2 + 7$.

- (a) Is particle X moving toward the left or toward the right at time $t = 1$? Give a reason for your answer.
- (b) At what time $t \geq 0$ is particle X farthest to the left? Justify your answer.
- (c) A second particle, Y , moves along the positive y -axis so that its position at time t is given by $y(t) = 7t + 3$. At any time t , $t \geq 0$, the origin and the positions of the particles X and Y are the vertices of a triangle in the first quadrant. Find the rate of change of the area of the triangle at time $t = 1$. Show the work that leads to your answer.

- (a) $x'(t) = 15t^2 - 18t$
 $x'(1) = 15 - 18 = -3$
 Since $x'(1) < 0$, the particle is moving to the left at time $t = 1$.

2 : $\begin{cases} 1 : \text{considers } x'(1) \\ 1 : \text{answer with reason} \end{cases}$

- (b) $x'(t) = 3t(5t - 6) = 0 \Rightarrow t = 0, t = \frac{6}{5}$

 Since $x'(t) < 0$ for $0 < t < \frac{6}{5}$ and $x'(t) > 0$ for $t > \frac{6}{5}$,
 the particle is farthest to the left at time $t = \frac{6}{5}$.

3 : $\begin{cases} 1 : \text{considers } x'(t) = 0 \\ 1 : \text{identifies } t = \frac{6}{5} \\ 1 : \text{answer with reason} \end{cases}$

- (c) Area = $A(t) = \frac{1}{2}x(t)y(t)$
 $= \frac{1}{2}(5t^3 - 9t^2 + 7)(7t + 3)$

 $A'(t) = \frac{1}{2}[(15t^2 - 18t)(7t + 3) + (5t^3 - 9t^2 + 7)(7)]$

 $A'(1) = \frac{1}{2}[(-3)(10) + (3)(7)] = \frac{1}{2}[-30 + 21] = -\frac{9}{2}$

4 : $\begin{cases} 1 : \text{area function} \\ 2 : \text{derivative} \\ 1 : \text{answer} \end{cases}$

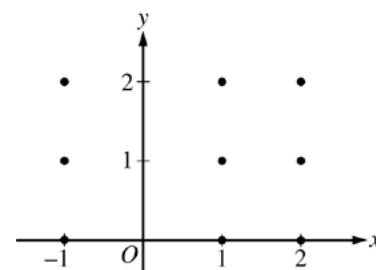
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Question 6

Consider the differential equation $\frac{dy}{dx} = \left(1 - \frac{2}{x^2}\right)(y - 1)$, where $x \neq 0$.

Let $y = f(x)$ be the particular solution to the differential equation with initial condition $f(1) = 2$.

- (a) Find the slope of the line tangent to the graph of f at the point $(1, 2)$.
- (b) On the axes provided, sketch a slope field for the given differential equation at the nine points indicated.
- (c) Find the particular solution $y = f(x)$ to the differential equation

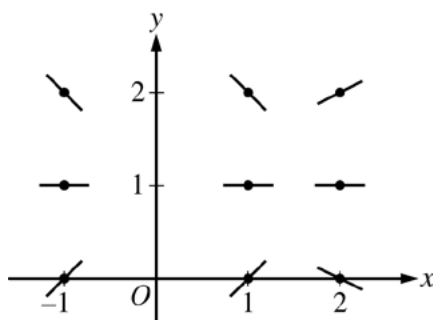


$$\frac{dy}{dx} = \left(1 - \frac{2}{x^2}\right)(y - 1) \text{ with initial condition } f(1) = 2.$$

(a) $\frac{dy}{dx}\bigg|_{(x,y)=(1,2)} = \left(1 - \frac{2}{1}\right)(2 - 1) = -1$

1 : answer

(b)



2 : $\begin{cases} 1 : \text{zero slope at each point } (x, y) \text{ where } y = 1 \\ 1 : \text{remaining slopes} \end{cases}$

(c) $\frac{dy}{dx} = \left(1 - \frac{2}{x^2}\right)(y - 1)$

$$\int \frac{dy}{y-1} = \int \left(1 - \frac{2}{x^2}\right) dx$$

$$\ln|y - 1| = x + \frac{2}{x} + C$$

$$\ln|2 - 1| = 1 + \frac{2}{1} + C \Rightarrow C = -3$$

$$\ln|y - 1| = x + \frac{2}{x} - 3$$

Note that $y - 1 > 0$ since the solution curve includes the point $(1, 2)$.

$$\ln(y - 1) = x + \frac{2}{x} - 3$$

$$y = f(x) = e^{\left(x + \frac{2}{x} - 3\right)} + 1$$

Note: This solution is valid for $x > 0$.

6 : $\begin{cases} 1 : \text{separation of variables} \\ 2 : \text{antiderivatives} \\ 1 : \text{constant of integration} \\ 1 : \text{uses initial condition} \\ 1 : \text{solves for } y \end{cases}$

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables